

The recovery research room

Teacher guide and worked answers

Use supplied sources and repeated sample data to present a cautious explanation of heart-rate recovery.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsg/inputs/GROW SOW 2026-27.xlsx | GROW Weekly - Spring | B36:C36

Research and present a healthy-body investigation.

Directly develops this stated outcome with secondary-age tasks at an upper-KS2 conceptual level. Support and stretch are flexible; this exemplar is not a claim of qualification approval.

Prior learning

Trace heart-lung-body circulation.

Read a table and subtract two whole numbers.

Distinguish a supported claim from a guarantee.

Success evidence

Select a relevant source and extract a useful scientific idea.

Describe a pattern using values from repeated data.

Present a claim, evidence, explanation and limitation without ranking health.

Equipment

Shared evidence and source summaries, selected route, pencils, optional calculator.

Preparation

Make clear that every number is constructed sample data, not a pupil or actual volunteer record.

No physical trial or online research is required. Print page 1 and the selected route.

Practical care

No exercise challenge, pulse measurement or comparison of pupils' bodies.

Do not interpret the fictional rates as diagnosis, fitness rankings or age-specific normal limits.

If adapting to a later real practical, the school must plan it separately for individual access and relevant health needs. This pack supplies a complete paper route.

Ready to teach alternative

Analyse supplied constructed records and paraphrased primary-source summaries. Present as a paper research briefing; no actual trial is needed.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Can we publish this headline?
2	Retrieval	2-5	Find the useful science
3	I do	5-13	I check what was compared
4	I do	Within stage	I describe a pattern using values
5	I do	Within stage	I check repeats before a headline
6	We do	13-21	Evidence desk: find the pattern
7	We do	Within stage	Which source helps explain it?
8	We do	Within stage	Write a better research headline
9	Hinge	21-24	A, B and C are three trials
10	You do	24-34	Prepare the research briefing
11	You do	Within stage	Use the reporting frame
12	You do	Within stage	Check your headline
13	Review	34-38	Research editor check
14	Review	Within stage	Choose the next evidence question
15	Exit	38-40	An honest research result

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Can we publish this headline?

Introduce a research briefing with no real pupil data. Pupils may challenge a headline without challenging another pupil's body or habits.

2 Retrieval Find the useful science

R1: blood. R2: working muscles have increased energy needs during activity and usually need more oxygen delivery. Accept "muscles working harder". R3: 96 beats per minute is a measurement with a unit; "fast" lacks a numerical reference.

3 I do I check what was compared

Think aloud: I cannot claim we measured classmates. The comparison represents one fictional person with three repeat trials. Set recovery times help comparison. The same activity and duration matter. Equal starting values were built into this dataset; real trials may have different starting values.

4 I do I describe a pattern using values

Shared I do time. Think aloud: $120 - 96 = 24$ beats per minute, the fall during the first minute. After two minutes, 80 remains 8 above the starting 72. The graph joins sample points to show a trend, not every measurement between them. This constructed pattern is consistent with recovery; it is not clinical proof.

5 I do I check repeats before a headline

Shared I do time. Model the mean: the total is 360, divided by 3 trials gives 120 beats per minute. In real trials, variation may reflect biological differences or measurement differences. These constructed values illustrate that idea. Do not delete 116 or 124 simply because they differ.

6 We do Evidence desk: find the pattern

All trials show a fall during recovery: A from 120 to 80; B from 116 to 76; C from 124 to 84 beats per minute over two minutes. Each final rate remains above the starting 72. Pupils choose a pair and explain the pattern. No physical exercise is needed.

7 We do Which source helps explain it?

Shared We do time. Source A is most directly relevant to oxygen delivery and heart-rate change. Source B provides broader activity context. Source C is an unattributed, unsupported absolute claim. The sources cannot prove that a constructed dataset came from a real trial or diagnose a person's health.

8 We do Write a better research headline

Shared We do time. Only one fictional person is represented; A, B and C are repeat trials, not three people. Model: In these sample trials, rates fell towards the starting value after activity. This cannot rank health or prove fitness.

Reveal: In these sample trials, heart rate fell during recovery. These records cannot rank health.

9 Hinge A, B and C are three trials

A is incorrect: these are three trials for one fictional person, not three people. B is the correct bounded pattern. C is incorrect: 80, 76 and 84 remain above 72. Revisit the labels before independent work.

A. Person B must be healthier than person A. - A and B label trials, not different people. Rates alone cannot rank health.

B. All three sample trials show heart rate falling during recovery. - Each trial falls from its immediate value towards its starting value.

C. Every trial is back to 72 after two minutes. - The final values are 80, 76 and 84, all above 72.

10 You do Prepare the research briefing

10 minutes across You do. Pupils use the supplied sources; no online access is needed. The briefing can be a paper panel, typed paragraph or dictated 60-second script.

11 You do Use the reporting frame

Shared You do time. The supported route uses trial A and source A. The standard route compares repeats. The stretch route calculates means and evaluates controls and causal claims.

12 You do Check your headline

Shared You do time. Require a constructed-data label near any chart or table pupils create. Do not accept faster recovery as proof of better health.

13 Review Research editor check

4 minutes across Review. A teacher or private check can replace peer review. Specific feedback: include values and units; distinguish a source explanation from an actual measurement.

14 Review Choose the next evidence question

Lundy: collect proposals and explain which method detail would be added to the next paper dataset. Accept more repeats, longer recovery recording, standardised timing or a clearer measurement method. This does not authorise an unplanned physical trial.

15 Exit An honest research result

E1: for example, trial A fell from 120 to 80 beats per minute over two minutes. E2: NHLBI for the heart-rate explanation. E3: constructed data for one fictional person cannot rank health. E4 on the pupil page asks for one control and its purpose.

Answers and assessment

R1-R3

R1: blood. R2: working muscles need more energy released in respiration, usually requiring more oxygen delivery. R3: 96 beats per minute is a measurement with units.

W1-W3

W1: all trials fall towards 72: A $120 \rightarrow 80$, B $116 \rightarrow 76$ and C $124 \rightarrow 84$ beats per minute from immediately after activity to two minutes later. Final values remain above 72. W2: source A, NHLBI, directly explains heart-rate response and oxygen demand; B provides general context; C lacks attribution, method and supporting evidence. W3: make a bounded claim about sample trials, not a health ranking. A, B and C are repeats for one fictional person.

Hinge A-C

A is incorrect: trial labels are not people and rates cannot rank health. B is correct: each trial shows a fall during recovery. C is incorrect: 80, 76 and 84 are not equal to 72.

S1-S5

S1: three repeat trials. S2: 120 to 80 beats per minute. S3: no, 80 is still above 72. S4: oxygen. S5 example: In this sample trial, heart rate fell during recovery. Trial A fell from 120 to 80 beats per minute over two minutes. These constructed records cannot diagnose or prove overall health.

N1-N4

N1: appropriate pairs from any two trials, showing a common fall but not a return to 72. N2: $120 - 96 = 24$ beats per minute. N3: source A, NHLBI; the advert has no author, method or supporting evidence and overclaims what one value can show. N4: bounded claim, accurate values and units, an oxygen-delivery explanation credited to NHLBI, and a limit such as constructed data, one fictional person, few repeats or limited recovery times.

T1-T4

T1: immediate mean = $(120 + 116 + 124) \div 3 = 120$ beats per minute; one-minute mean = $(96 + 92 + 100) \div 3 = 96$ beats per minute. T2: the mean falls by 24 beats per minute. This does not establish a universal response, especially from constructed data for one person. T3: any two controls among activity, duration, room, recovery posture and measurement timing or method; each reduces an avoidable comparison difference. A useful improvement is more trials to explore variation, longer follow-up, or a clearer standardised measurement procedure. T4: report three sample trials for one fictional person, an accurate recovery pattern, numerical evidence, explanation and a limit.

E1-E4 / reflection

E1: accurate paired values and units. E2: NHLBI for the heart-rate and oxygen-delivery explanation; NHS is acceptable for broad activity context if relevance is clear. E3: no clinical diagnosis, health ranking, claim of real measurement or universal conclusion. E4: one genuine control with an explanation of its comparison purpose. Reflection accepts a precise question for the next paper dataset.

Assessment and re-teach

Secure: uses accurate values and units, identifies a relevant source, explains a heart-rate response and states an appropriate limit. If a health ranking appears, circle the trial labels and ask how many people are represented. If a pattern lacks numbers, require one paired comparison. If a mean is confused with a total, model total divided by number of trials.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 8; We do 8; hinge 3; You do 10; review 4; exit 2. Zero-minute slides share their stage time.

Supply shared page 1, one selected route page 2, 3 or 4, and page 5. Do not require all three routes. An independent spoken explanation can be recorded verbatim by an adult.

The research task is self-contained: pupils compare primary-source summaries, analyse constructed data and present findings. It meets the SoW's research/presentation focus without requiring internet accounts or any physical experiment.

All timing, rates and repeat methods are authored examples. They are not a recommended exercise prescription and not age-specific clinical reference values.

Interactive predictions map to the paper method/data/source tasks W1-W3. A pupil can test the claim by tracing the printed table and explaining the same conclusion.

Access and responsive teaching

Supported

Read one trial; choose a relevant source and use a briefing frame.

Standard

Compare repeats and present claim/evidence/explanation/limit.

Stretch

Calculate selected means and evaluate a causal overclaim and test design.

Regulation

Offer quiet pointing, private writing, drawing or adult scribing. Pass-and-return for public questions. Use fictional cases; no personal medical disclosure or comparison of bodies. Assign response routes from current evidence, not fixed pupil labels.

Misconceptions to check

A single rate proves how healthy somebody is.

Many influences affect heart rate; this dataset cannot diagnose or rank health.

Check: Ask what the records tell us and what they leave unknown.

One repeat must exactly match the others.

Repeated measurements can vary; variation helps us judge confidence.

Check: Compare immediate values 120, 116 and 124.

A source about general benefits proves this small trial's conclusion.

A relevant source can explain a pattern but does not turn constructed sample data into a real study.

Check: Ask whether source statements and the sample measurements are the same kind of evidence.

Word	Meaning
heart rate	The number of heartbeats per minute.
recovery	The period after an activity while body responses move back towards a resting state.
repeat	Another trial using the same planned method.
variation	Differences between measurements or trials.
mean	The total of the values divided by the number of values.
limitation	A reason to restrict how far a conclusion can reach.

Scientific references

NHLBI: How the heart beats

Checked September 2026. During exercise heart rate usually increases to support oxygen delivery. No adult normal-range figures are applied to pupils.

NHS: Physical activity for children and young people

Checked September 2026. General activity context; the classroom investigates supplied constructed data, not pupils' fitness or medical status.

Department for Education: Science programmes of study

Working-scientifically framing: evidence, reporting and conclusions. Activities are original and follow the user's SoW.